Facial Emotion Recognition Using Masked-Augmented FERPlus

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***Abstract*— The Masked-Augmented FERPlus dataset seeks to resolve the issue of dataset imbalance in emotion detection, particularly when faces are obstructed by objects such as masks. The objective is to determine the appropriate oversample size using the classification complexity metric to address the dataset imbalance in the original FERPlus dataset. The dataset was used to train various machine learning models, including Convolutional Neural Networks (CNN), VGG16, ResNet, and InceptionV3, and their accuracy was measured to determine the most effective model. Although the oversampled dataset demonstrated superior performance with the N1 metric calculation, the model's accuracy on the original dataset, which was 0.752, slightly surpassed its performance on the masked augmented dataset, which achieved an accuracy of 0.747. Despite the simple augmentation technique implemented in this study, the results indicated that it effectively reduced model overfitting while training.**

Keywords—emotion recognition, data augmentation, Convolutional Neural Network, classification complexity, FERPlus

I. Introduction

Human emotion detection is a subject that has been actively explored by researchers [1]. Due to the Covid-19, the education system in Indonesia has gone through significant changes, with schools relying heavily on online media as the primary tools for teaching and learning [2]. Hence, there is now an urgent need for an emotion detection model that can detect emotions even when the face is partially or entirely obstructed by a mask or other objects.

This study aim to develop emotion detection model when face obstructed by objects. To achieve this goal, a dataset containing human emotions with obstructions is necessary. Existing public datasets, like FER2013, CK+, and JAFFE [3], can be utilized. In this research, we introduce the Masked-Augmented FERPlus dataset, an updated version of FER2013 originally released by Microsoft. However, this dataset has an unbalanced distribution. By employing FERPlus augmentation based on classification complexity, we were able to enhance the sample size while maintaining balance [4]. After augmenting the dataset, we added face masks to address the issue of emotion detection when wearing masks. We assess the accuracy of various machine learning models, such as the fundamental CNN, Resnet, VGG16, and InceptionV3, using the Masked-Augmented FERPlus dataset.

II. Related Work

In a study by Felipe Zago Canal et al. [5], the researchers delineated five key stages in constructing a Face Emotion Recognition system, comprising face input, image preparation, feature selection, classification, and validation. The authors assert that the formulation and implementation of algorithms for human emotion detection create novel prospects for human-computer interaction, including robotics, intelligent tutoring systems, and digital marketing.

Numerous studies in the field of Facial Emotion Recognition have resulted in the creation of new datasets. In a comprehensive literature review, Prameela Naga et al. [3] have summarized several commonly used datasets in building facial emotion recognition models. A detailed description of these datasets is presented in Table 1. Additionally, Xu et al. [6] conducted a study in which they developed a dataset called the Spontaneous Visual Database for Online Learning (SVD-OL). This dataset classifies students' emotions into six categories: interested, confused, frustrated, happy, sleepy, and neutral. The model built in this study employs a Convolutional Neural Network (CNN) and achieves an accuracy of 79%.

TABLE I. FACIAL EMOTION DATASETS [3]

Name	Description	Size	Type
JAFFE	213 poses of Japanese women labeled with 7 basic emotions plus neutral	1,000	Image
FER2013	Images represented with pixels and labeled 0-6 based on human emotions	35,887	Image
CK+	Various photos of 18-30 year old humans of African-American, Asian and Latino races	500	Image
CMU-MultiPIE	Images of the subject taken in 19 different lighting situations	750,000	Image
AffectNet	Emphasis on valences that identify images in a continuous dimensional model	1,000,000	Image
IEMOCAP	Addition of emotion levels based on the situation	12 hours	Video
RAF-DB	30,000 face images in the crowd	30,000	Image
CAER	Selected video clips to identify emotions	13,000	Video
iCV-MEFED	Compound emotions	31,250	Image
AFEW	Dynamic facial expressions taken from the movie	-	Image
KDEF	Photos taken at 5 different angles	4,900	Image

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CNN-based models have been extensively utilized in several research studies. For instance, Huang et al. applied CNN models in constructing facial emotion recognition on the RAF-DB and AffectNet datasets [7]. In another study, researchers combined CNN with the YOLO algorithm to detect human faces and classify their emotions [8]. Furthermore, Shahzad's research involved altering the classifier at each layer of a neural network (CNN) using a Support Vector Machine (SVM) [9].

Another study was carried out by Mike Rinck et al. [10] with the aim of identifying human emotions in the presence of obstructive objects on the face, such as masks. The researchers utilized the Radboud Faces Database as the dataset for this study. The findings of this research indicated that the model's accuracy dropped by approximately 20% when the face was obstructed by an object. The misclassification of emotions was most prominent in the cases of disgust, fear, surprise, sadness, and happiness.

Imane Lasri et al. [11] developed a student emotion detection system using a Convolutional Neural Network (CNN) in their research, which could aid teachers in modifying their teaching methods based on student emotions. The FER2013 dataset was utilized in this study. The proposed system comprised three main stages. The initial stage involved face detection through the Haar cascade method. The second stage entailed cropping and normalizing the image. The final stage involved emotion recognition using CNN. The CNN model employed comprised four convolutional layers, four pooling layers, and two fully connected layers. This study achieved an accuracy of up to 70%.

In the year 2023, Bin Han et al. [12] undertook research aimed at expanding the FER2013 dataset and incorporating masks into the dataset to recognize a person's emotions while wearing a mask. Multiple models were utilized to evaluate the effectiveness of the Masked FER2013 dataset. The table below presents the accuracy levels of the various models that were examined in this study.

TABLE II. COMPARISON OF MODELS ACCURACY TRAINED ON FER2013 AND MASKED FER2013 [12]

	FER2013	**Masked FER2013**
CNN	30%	43%
VGG 19	42.5%	62.42%
ResNet 50	18%	64.94%

Improving the quality of a dataset is an important step that researchers can take, and one such method is data augmentation. This can involve adjusting the contrast and brightness of images, as well as resizing and applying filters to them [3]. While numerous studies have focused on developing or enhancing specific datasets related to human emotions, there is a lack of research that defines the optimal level of augmentation required for unbalanced datasets to prevent overfitting during training [13]. To address this issue, Dohyun Lee and Kyoungok Kim conducted a study that proposed a novel approach. This approach involves determining the classification complexity and using N1 as a variable to establish the size of the class in the dataset. N1 calculates the ratio of distances between samples within the same class (intra-class) compared to the distances between samples in other classes (extra-class).

III. METHODOLOGY

There are two major steps in this research, namely dataset creation and modeling. We can see the overview of the research steps in Figure 1. The followings are an explanation of each step in this research.

Figure 1, Part A displays the initial stage in this research, which is the creation of the dataset. This study evaluates two different approaches to address the imbalanced class issue in the FERPlus dataset. The main objective of this research is to determine the optimal oversampling size for the dataset. The first approach involves increasing the number of images to the maximum extent possible, while the second approach takes into account the complexity of the classification task to determine the appropriate oversampling size. The augmentation technique employed in this study is a straightforward approach of rotation, stretching, and shifting using the Keras ImageDataGenerator library.

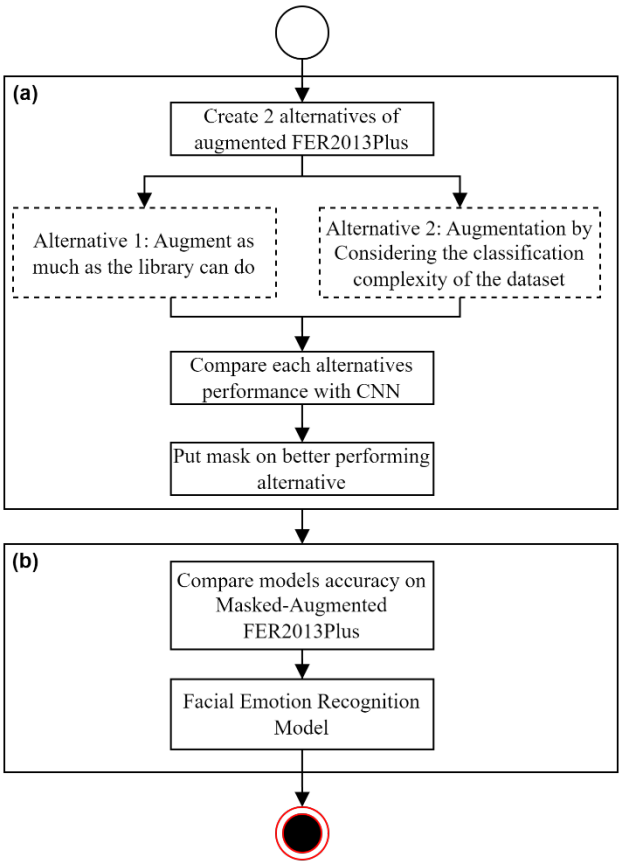

Fig. 1. Overview of this research steps. (a) Diagram of dataset creation step; (b) Diagram of modeling step

Following the creation of two alternative datasets for FERPlus, we proceeded to compare their performance using a CNN-based model, as depicted in Figure 2. Upon identifying the dataset with superior performance, we incorporated masks into the images by employing a straightforward computer vision script. Specifically, we added masked faces to the dataset, equivalent to 25% of the size of each class, to maintain balance in the dataset.

After generating Masked-Augmented FERPlus, we continued experimenting to generate the best inference for

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facial emotion recognition. There are various models that will be tried and compared for accuracy ranging from CNN-based models whose architecture can be seen in Figure 3, VGG16, ResNet, and IncetionV3. The accuracy evaluated is the validation accuracy during training. All training utilizes cloud services to speed up training and reduce computing load on local computers.

IV. Experiment

A. Dataset Creation

The base dataset used in this research is FERPlus, a dataset that is an update of FER2013 launched by Microsoft. The dataset is publicly available and can be accessed via *github.com/Microsoft/FERPlus*. The difference between FERPlus and FER2013 is that there are some images that are relabeled based on more appropriate classes. Figure 2a shows the class imbalance of the FERplus dataset.

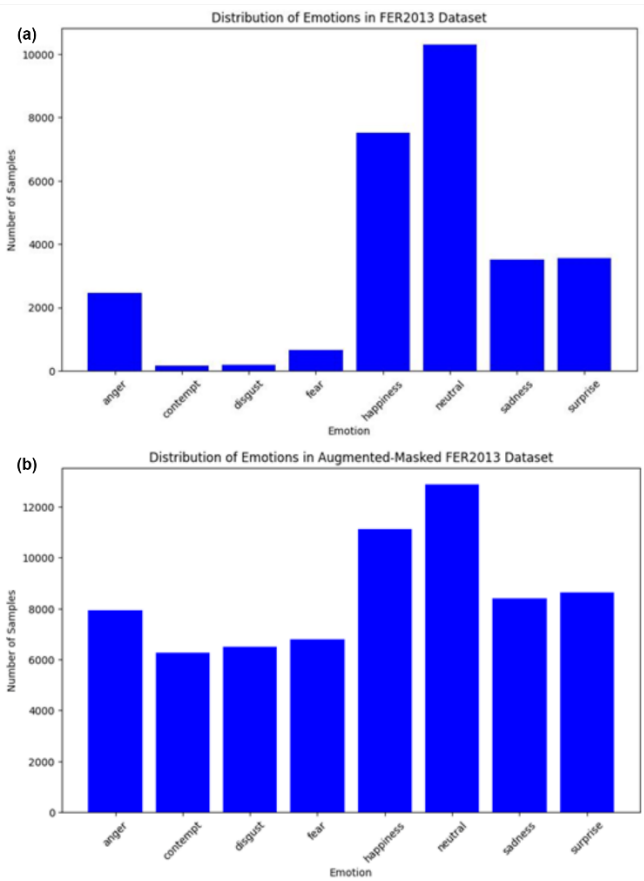

Fig. 2. Overview of the comparison of the number of pictures of each class: (a) Number of images of each class in the original FERPlus dataset; (b) Number of images of each class in the Masked-Augmented FERPlus dataset

TABLE III. Keras ImageDataGenerator Configuration

rotation_range	**30**
width_shift_range	0.1
height_shift_range	0.1
shear_range	0.1
zoom_range	0.1
horizontal_flip	True
fill_mode	'nearest'

The use of data augmentation techniques can be effective in addressing the issue of class imbalance in the FERPlus dataset. In this study, a straightforward augmentation technique was employed in the form of rotation, stretching, and shifting, utilizing the Keras ImageDataGenerator library. The specific configuration of the ImageDataGenerator can be found in Table 3. Two augmentation alternatives were explored for each class: 1) performing as many augmentations as possible, and 2) performing a number of augmentations based on the calculation of classification complexity as described in [13].

1) Oversampling Size Determination

Alternative 1 does not involve calculating the oversample size and relies solely on the library for assistance. Conversely, in the second alternative, we utilize the classification complexity metric to determine the appropriate oversample size. The N1 metric is the type of metric we employ, which calculates the ratio of distances between samples within the same class (intra-class) and distances between samples in other classes (extra-class). The calculation of distances between samples in the same class and different classes is depicted in equation (1).

$$intra_extra = \frac{\sum_{i=1}^{n} \sum_{j \in NN1,intra(X_i)} d(X_i, X_j)}{\sum_{i=1}^{n} \sum_{j \in NN1,extra(X_i)} d(X_i, X_j)} \quad (1)$$

The calculation of the N1 metric involves utilizing the distance measurements between samples within the same class and between samples from different classes, as demonstrated in equation (2).

$$N1 = 1 - \frac{1}{1+intra_extra} = \frac{intra_extra}{1+intra_extra} \quad (2)$$

After getting the N1 metric from (2), we use it to determine the oversampling size.

$$O = \left(|C_n| - |C_p|\right) \times N_1, N_1 \in [0,1] \quad (3)$$

Where O is oversample size, Cn is the negative (majority) class, and Cp is the positive (minority) class. In the FERPlus dataset, the minority class is 'contempt' while the majority class is 'neutral'. The size of O is limited to not exceed Cn to fulfill the purpose of this calculation, which is to get a sufficient and optimal oversampling number to avoid overfitting when training models.

TABLE IV. Number of Images of FERPlus, Augmentation 1, Augmentation 2

	Original	Augmentation 1	Augmentation 2
Anger	2466	9952	6347
Contempt	165	2928	5184
Disgust	191	3278	5198
Fear	652	7508	5431
Happiness	7528	10000	8904
Neutral	10308	10000	10308
Sadness	3514	9993	6876
Surprise	3562	9995	6901

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Table 4 presents the count of images from each class in the original FERPlus dataset, alternative augmentation 1, and alternative augmentation 2. Following the testing of the CNN model depicted in Figure 3, it was determined that alternative 2 achieved a superior accuracy of up to 10% compared to alternative 1. As a result, alternative 2 incorporates 25% masked face images for each class to ensure balance between classes. This addition was accomplished using a straightforward computer vision script that can be accessed at *github.com/aqeelanwar/MaskTheFace*.

B. Modeling

The development of a Convolutional Neural Network-based model was undertaken to assess both Augmented FERPlus alternatives. The architecture of the CNN model is depicted in Figure 3. Following the determination of the FERPlus augmented dataset that exhibits superior performance, the dataset was evaluated on three pre-trained models, namely VGG16, ResNet, and InceptionV3. Subsequently, the four architectures were compared with respect to accuracy on the Augmented-Masked FERPlus dataset, with the aim of identifying the best model with the highest accuracy.

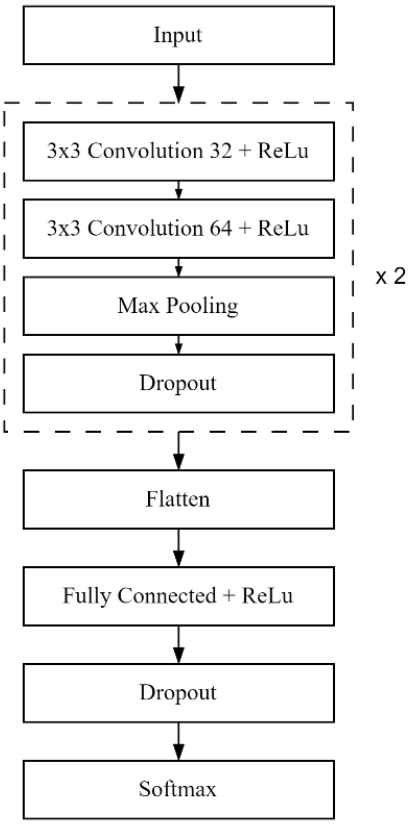

Fig. 3. Overview of the CNN architecure that will be used

V. Result and Analysis

The following is a more detailed explanation of FERPlus augmentation and the experimental results and accuracy of various models.

A. Masked-Augmented FERPlus

Based on the calculation derived from (3), Table 2 presents a comparison of the number of images in each of the eight emotion classes, which include anger, contempt, disgust, fear, happiness, neutral, sadness, and surprise.

The evaluation of the two augmentation methods involves utilizing the CNN-based model depicted in Figure 2. The first method entails applying the maximum amount of augmentation feasible, based on the capabilities of the library, and achieving an accuracy of 0.6626. The second method, which employs augmentation based on the calculation of classification complexity, achieves an accuracy of 0.7458.

In light of the higher accuracy attained through the second alternative, we employed a computer vision script to apply the mask to the augmented dataset. To maintain balance within the dataset, we added approximately 25% masked face images to each class. As depicted in Figure 4, an example of each emotion class that has been masked is provided. As illustrated in Figure 2b, the final distribution of our Augmented-Masked FERPlus dataset is displayed.

Fig. 4. Examples of our Masked-Augmented FERPlus dataset. The first row are the augmented example, while the second row are the masked version of each augmented picture

B. Models Accuracy

We conducted a comparison of the performance and accuracy of several models on our Masked-Augmented FERPlus dataset. Specifically, we tested the basic CNN-based model depicted in Figure 3, as well as VGG16, ResNet, and InceptionV3. Our study involved two main hyperparameter configurations, which are presented in Table 5.

TABLE V. Hyperparameter Configuration for Training Models

	Configuration 1	Configuration 2
epochs	50	100
batch_size	32	64
learning_rate	0.001	0.0001

After setting the hyperparameters for training, we gathered accuracy data for each model. The training was conducted entirely on separate servers. For each model, we conducted six training iterations, covering three datasets and two configurations. Table 6 presents the accuracy metrics for each model and configuration discussed earlier.

TABLE VI. MODELS ACCURACY ON FERPLUS, AUGMENTED FERPLUS, AND MASKED-AUGMENTED FERPLUS

	Conf	CNN	VGG16	ResNet	InceptionV3
FERPlus	1	0.754	0.533	0.464	0.530
	2	0.752	0.519	0.442	0.531
Augmented	1	0.723	0.490	0.405	0.490
	2	0.746	0.478	0.371	0.485
Masked-Augmented	1	0.366	0.480	0.412	0.478
	2	0.747	0.464	0.371	0.480

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Based on the results obtained, it can be seen that the model that has the best performance and is able to achieve the highest accuracy is the CNN-based model found in figure. Although the CNN-based model is able to get the highest accuracy, but when viewed further, the performance of the original FERPlus dataset model is slightly better than the augmented dataset. The following are the training details for the CNN-based model on the augmented and augmented-masked datasets.

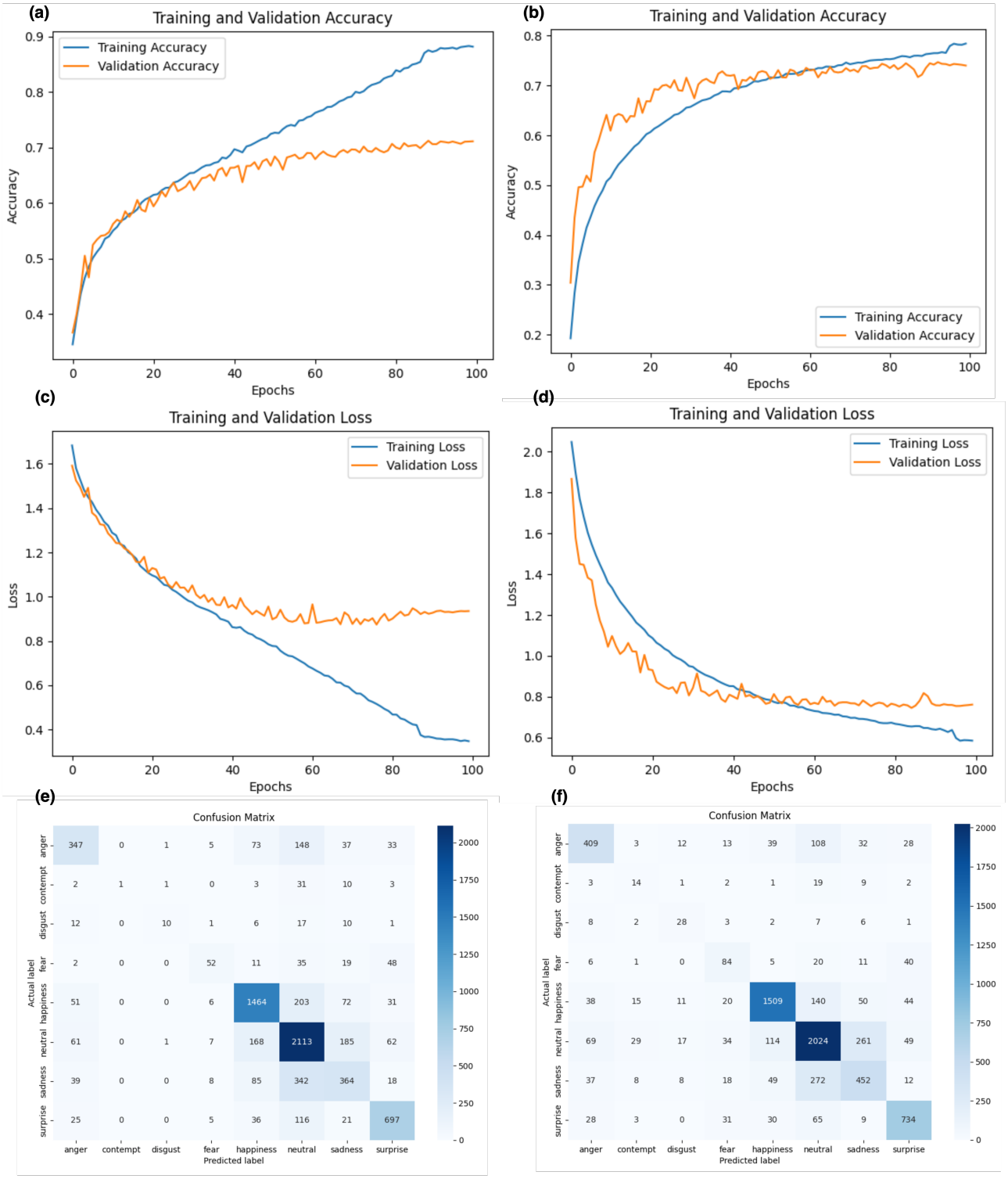

Fig. 5. Detail of training for CNN based model on FERPlus and Masked-Augmented FERPlus dataset with configuration 2. (a) CNN accuracy on FERPlus; (b) CNN accuracy on Masked-Augmented FERPlus; (c) CNN training loss on FERPlus; (d) CNN training loss on Masked-Augmented FERPlus; (e) Confusion matrix on FERPlus; (f) Confusion matrix on Masked-Augmented FERPlus.

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Figure 5a indicates that the model experienced overfitting during training on the FERPlus dataset. In contrast, the model performs better when trained on the Augmented-Masked FERPlus dataset, as evidenced in Figure 5b. Despite this improvement, the accuracy achieved by the CNN when trained on the FERPlus dataset is higher, as illustrated in Table VI.

Similarly, the training loss for the model appears to be affected by overfitting when training on the FERPlus dataset, as depicted in Figure 5c. The loss reduction for the CNN trained on the Augmented-Masked FERPlus dataset, however, is more noticeable compared to the CNN trained on the FERPlus dataset, as shown in Figure 5b.

Therefore, it can be concluded that although the model trained on the Augmented-Masked FERPlus dataset exhibits a slight reduction in accuracy compared to the model trained on the original FERPlus dataset, as demonstrated in Table VI, the Augmented-Masked FERPlus dataset effectively reduces overfitting during training. Moreover, the minor reduction in accuracy displayed by the model on the Augmented and Masked-Augmented datasets is not due to architectural flaws in the model but rather to the datasets themselves. Notably, the performance of the second augmented dataset alternative is superior to the first, suggesting that the size of the oversampling applied to the FERPlus dataset is appropriate. However, the technique used to augment the FERPlus dataset requires improvement as it only involves simple rotations, stretching, and shifting, which is insufficient in enhancing the model's ability to recognize new images that have not been previously seen.

In addition, when examining the confusion matrix in Figures 5e and 5f, it is evident that the model experiences the greatest difficulty in classifying negative emotions, such as contempt, disgust, and fear. On the other hand, the model performs remarkably well in identifying the emotions of anger, happiness, neutrality, sadness, and surprise.

VI. Conclusion

The following study introduces the Masked-Augmented FERPlus dataset, which is derived from FERPlus. The aim of this dataset is to tackle the issue of imbalanced data in FERPlus and to enable the detection of human emotions even when obstructed by objects, such as masks. The approach taken to determine the oversampling size is based on the classification complexity of the dataset, and it has demonstrated improved performance of up to 10% as compared to oversampling sizes that have not been calculated.

The Masked-Augmented FERPlus dataset was subsequently subjected to training using various models, and the outcomes indicate that the Convolutional Neural Network (CNN) exhibits superior performance compared to other models, achieving an accuracy of 0.746 for the Augmented FERPlus and 0.747 for the Masked-Augmented FERPlus datasets. Although the results obtained from the CNN model trained on the original FERPlus dataset were slightly higher than those obtained from the Augmented-Masked FERPlus dataset, the latter was found to reduce overfitting during training. The slight decrease in accuracy can be attributed to the augmentation technique used, which involves simple rotation, stretching, and shifting, causing the model to perform less effectively on unseen images.

Furthermore, the outcomes of this study indicate that the majority of incorrect emotional classifications were linked to negative emotions, including contempt, disgust, and fear. However, the model has proven to be quite effective in recognizing the emotions of anger, happiness, neutrality, sadness, and surprise.

In our future endeavors, we envision enhancing and refining the augmentation methodologies we employ. To achieve this objective, we intend to explore the potential of Generative Adversarial Network (GAN) technology, with the aim of exposing the model to a greater variety of images during its training phase. This approach is expected to yield superior performance compared to previous iterations.

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